Separation of Church and State Curricula? Public Standards, Private Values, and Textbook Content

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Working Paper, 2026

Prepared for ifo Summer Course 2026, by Benjamin W. Arold

Motivation & Research Question

- Curricula are a critical site of cultural transmission: they convey knowledge, instill values, and shape children's worldviews - with causal effects on later beliefs, religiosity, and careers.
- U.S. school-choice programs increasingly direct tax dollars to religious private schools and homeschooling, which face minimal curricular oversight - yet we know little about WHAT they teach.
- Question: How do the people, topics, and values in textbooks differ across public (Texas, California) vs. religious private/home-school settings, and over time?
    - Two puzzles: Why are 'red' Texas and 'blue' California textbooks so SIMILAR, while religious books DIVERGE - despite the polarization narrative?
    - New in this version: a market/regulatory framework that explains this pattern, tested via cross-state spillovers from standards revisions.

Data: A Novel Panel of 261 Textbooks (1980-2022)

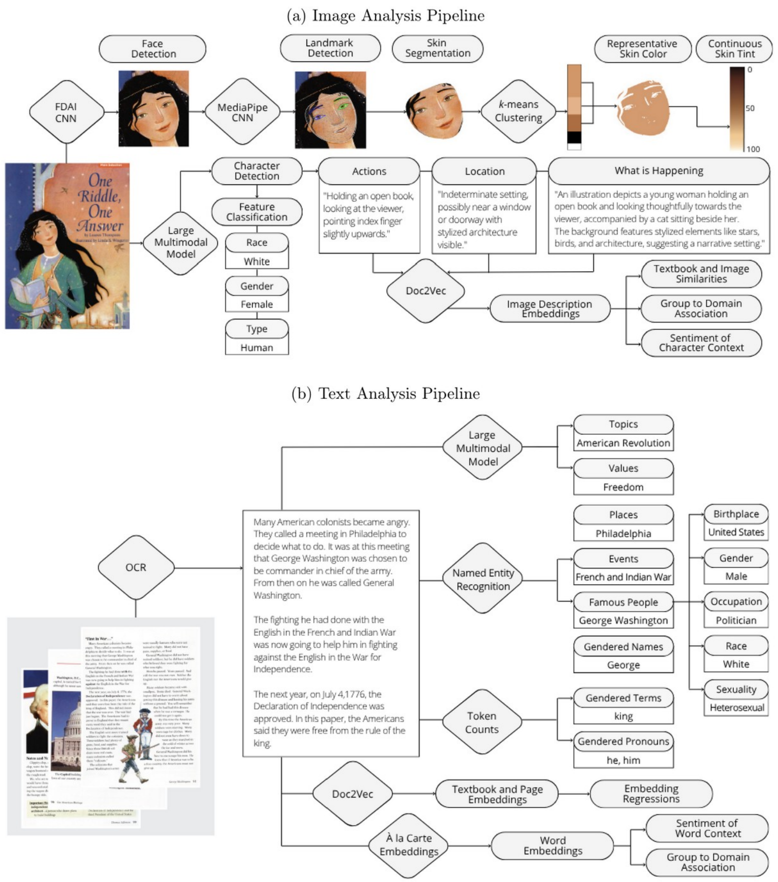
- Three collections: Texas public, California public, and Religious (conservative Christian publishers Abeka, BJU Press, and ACE - widely used in private and home schools).
- Subjects: Reading, Science, Social Studies; grades 3 and 5 (primary-school-age children).
- Scale: 261 digitized textbooks, ~88,000 pages, ~16 million words, ~60,000 detected faces.
- Longitudinal, 1980-2022: enables comparisons across settings AND over four decades.
- Texas and California are the two largest K-12 markets and the conservative/liberal poles of public education; religious publishers serve over a third of taxpayer-funded non-Catholic Christian schools.

Computational Methods: What, Who, and How

- Each scanned page is treated as BOTH an image and a source of text, analyzed with AI + computational social-science tools at scale.
- WHAT is communicated: topics and values on each page (LLM + keyword validation).
- WHO is depicted: representation by race, gender, skin color, sexuality, birthplace, occupation (computer vision + NER).
- HOW people are portrayed: the roles, settings, actions, and sentiment surrounding each group (embeddings + VAD).
- Lets one study measure both PRESENCE and PORTRAYAL across settings and decades - replicating prior qualitative work while uncovering new patterns.

Method Pipeline: Turning Images and Text into Data

Figure 1: Data Extraction Pipelines: Image and Text Analysis



Note: Image pipeline (top): a Large Multimodal Model (Gemini 2.5 Pro) detects characters, actions, setting, race/gender/type; a CNN face detector (FDAI) + MediaPipe landmarks feed skin segmentation and k-means clustering to a continuous skin tint (0=darkest, 100=lightest); Doc2Vec embeds image descriptions. Text pipeline (bottom): OCR feeds a GPT-4 LLM (topics/values), Named Entity Recognition (places, events, famous people), token counts of gendered terms, and page/word (a-la-carte) embeddings. (Figure 1)

Computational Methods in Depth

- Images (LMM): Gemini 2.5 Pro detects characters (incl. faces not visible & non-human), their race, gender, actions and setting; outperforms the CNN classifier used for robustness.
- Skin color (computer vision): CNN-detected faces are skin-segmented, k-means clustered, and scored on the L^* axis of CIE $L^*a^*b^*$ space (0=darkest to 100=lightest).
- LLM (GPT-4): free-form annotation of topics and values per page (hypothesis-generating, not a fixed list); validated vs. human labels and keyword counts.
- NER + Pantheon 2.0: identify places, events, and famous people, linked to gender, occupation, sexuality, birthplace; race manually coded; SSA name data for gender.
- Embeddings: a-la-carte (conText) word embeddings + Doc2Vec page/book/image-description embeddings; normed embedding regressions test differences across collections.
- Sentiment: NRC VAD lexicon scores Valence, Arousal, Dominance of the words most associated with each gender, in both text and images.

Headline: Texas ~ California; Religious Books Stand Apart

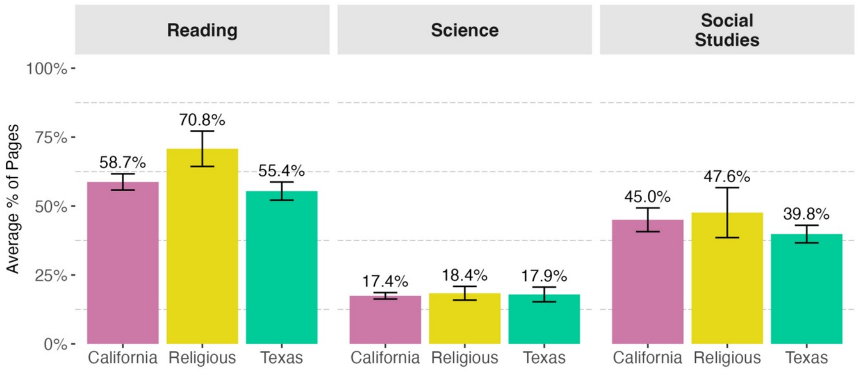
Figure 2: Textbook Embeddings

Note: t-SNE maps of book-level TEXT (a) and IMAGE-description (b) embeddings: books cluster by subject, and within subject California and Texas overlap while Religious separates. Panels (c)-(d): normed coefficients from embedding regressions. The California-vs-Texas difference is small and NOT significant (0.04, p=0.21 text / 0.23 image), whereas Religious-vs-Texas and Religious-vs-California gaps are larger and highly significant. Contrary to the polarization narrative, the two public systems teach very similar content. (Figure 2)

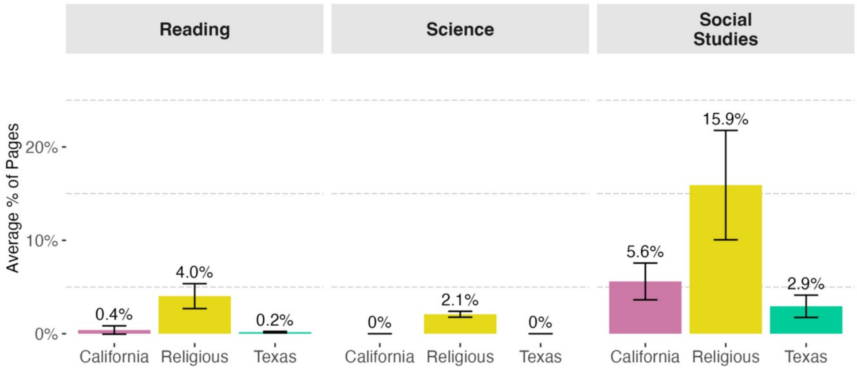
Values: Religious Books Emphasize Religious & Character Values

Figure 3: Values

(a) Average Percent of Pages Containing Character Values by Subject



(b) Average Percent of Pages Containing Religious Values by Subject



Note: Share of pages containing character values (a) and religious values (b), by subject and collection. Character values are pervasive everywhere (Science 17-18%, Social Studies 40-48%), but Religious Reading books are highest (71%). Religious values are far more prevalent in the Religious collection across ALL subjects - 15.9% of Social Studies pages vs. 5.6% (CA) and 2.9% (TX), and 2.1% even in Science where public books are near zero. Texas and California show no meaningful difference. (Figure 3)

NEW Framework: Market & Regulatory Forces Drive Content

- Publishers serve markets; content is chosen like a product position (Hotelling/Downs). The question: why converge in public, diverge in religious?
- Public-school publishers (e.g. Pearson, McGraw Hill) serve a NATIONAL market: they build one baseline 'national' edition and only customize per state when sales justify the fixed cost F.
  - State standards restrict the feasible set; with high customization costs, the baseline converges to the 'safe union' of topics required in some state and banned in none.
  - Prediction 1 - positive SPILLOVERS: a topic required by one state appears in other states' editions too. Prediction 2 - CHILLING: topics banned by an influential state get dropped from the baseline.
- Religious publishers serve a SEGMENTED, less-regulated market not bound by state standards -> free to DIFFERENTIATE toward their target market's demographics, beliefs, and values.
- Policy implication: expanding public funds for private schooling without shared standards could increase curricular - and long-run belief - divergence.

Evidence: Standards Revisions Spill Over Across States

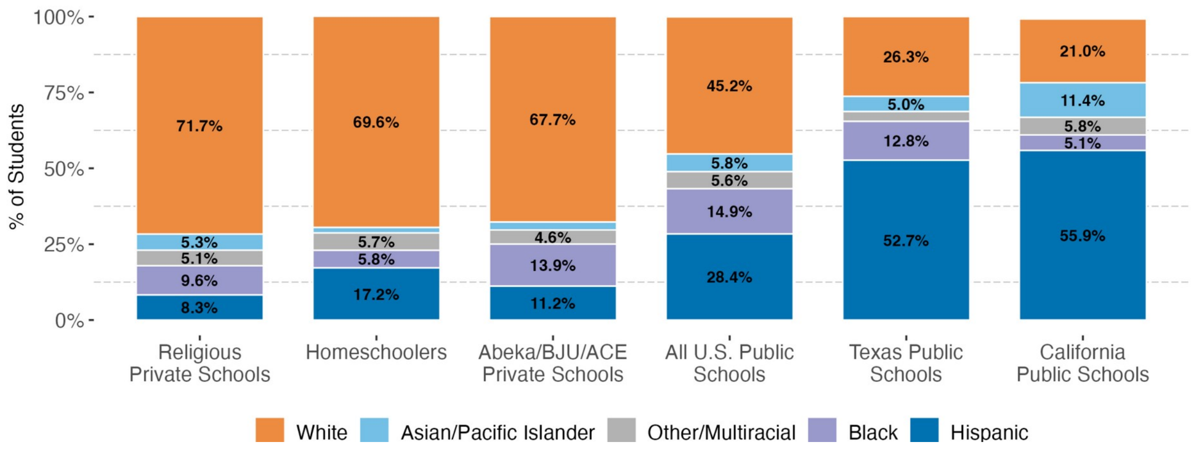
Table 2: Direct Effects and Cross-State Spillovers from State Standards Revisions

	Dependent Variable: % New-Standards Words			
	All Subjects	Reading	Science	Social Studies
	(1)	(2)	(3)	(4)
Direct Effect	0.0160***	0.0101***	0.0244***	0.0042**
	(0.0020)	(0.0018)	(0.0031)	(0.0017)
Spillover	0.0023***	0.0018*	0.0026	0.0027**
	(0.0008)	(0.0010)	(0.0021)	(0.0011)
Fixed-effects				
Collection-Grade-Subject	Yes			
Collection-Grade		Yes	Yes	Yes
Observations	735	285	271	179
Within R^2	0.49186	0.39748	0.67932	0.35463

*Note: Textbook-year regressions of the share of 'new-standards' words (words new to the revising state's standards) on post-adoption indicators, with collection-grade(-subject) fixed effects. Direct effect: the revising state's books adopt ~16 new-standards words per 1,000 (0.0160). Spillover: the OTHER state's books also rise (~2.3 per 1,000; 0.0023***) despite no change to ITS standards - positive for Reading and Social Studies. No Science spillover, consistent with divergent Science standards (California adopted NGSS in 2013; Texas did not). (Table 2)*

Differentiation Aligns with Target-Market Preferences

Figure 7: Racial/Ethnic Composition of K–12 Students by School Type



Note: Racial/ethnic composition of K-12 students by school type (NCES). Religious private, homeschool, and Abeka/BJU/ACE populations are ~68-72% White. (Figure 7)

- Religious/homeschool students are ~70% White vs. ~21% (CA) and ~26% (TX) public - a demographic gap that rationalizes lighter, whiter imagery.
- Consistent with this, Religious books depict lighter skin (even conditioning on race) and more White people in text and images.
- GSS beliefs of religious respondents track the content: more rejection of evolution, support for school prayer, opposition to homosexuality, and traditional gender norms (Fig. 8).
- Portrayal (VAD, Fig. 6): across ALL collections females are framed more positively but less active and less powerful than males; Religious books have the lowest

Takeaways

- No deep public-school divide: despite the 'red vs. blue' narrative, Texas and California textbooks are strikingly similar in topics, values, portrayal, and representation.
- Church vs. state IS visible: religious books emphasize religious and character values, show less female representation, and depict lighter-skinned, more White figures - including in Science.
- A market explanation: state standards + customization costs push public publishers toward a converged national baseline (with measurable cross-state spillovers); unregulated religious publishers differentiate toward their target market.
- Shared blind spots everywhere: minimal LGBTQIA+, Asian, Latine, and Indigenous representation; females portrayed positively but less powerfully.
- Methodological contribution: a scalable AI + text/image pipeline (LMM, LLM, computer-vision skin analysis, NER, embeddings, VAD) for measuring presence and portrayal in any large document corpus.